SMART COMMUNITY-BASED WOMEN PROTECTION AND INCIDENT REPORTING PLATFORM

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ABSTRACT

This study investigates and develops SafeCity, a smart community-based women protection and proactive incident reporting platform engineered to overcome the systemic limitations of disconnected mobile safety tools and reactive panic buttons. In contemporary urban centers, over 70% of street-level harassment, stalking, and eve-teasing incidents go unrecorded due to social stigma, fear of retaliation, and reporting friction—a phenomenon criminologists define as the 'dark figure of crime.' Furthermore, mainstream navigation systems systematically prioritize shortest travel distance over personal safety, routing pedestrians through hazardous, unlit transit corridors. To resolve these challenges, SafeCity integrates an anonymous, privacy-preserving incident reporting framework with real-time geospatial visualization, multi-criteria safety-aware route optimization, a live GPS virtual companion escort, and an intelligent conversational guide. Evaluated across metropolitan transit corridors in Bengaluru, empirical results demonstrate sub-second route calculation (median 320 ms), robust 12-meter GPS tracking accuracy, a high-precision 180-second inactivity watchdog, and significant safety discrimination (94/100 for illuminated arterials vs. 48/100 for hazardous shortcuts). This work demonstrates that combining grassroots participatory reporting with multi-objective navigation algorithms transforms passive citizen accounts into proactive, life-saving civic mobility intelligence.

KEYWORDS: *Women's Safety, Community-Based Incident Reporting, Geographic Information Systems (GIS), Location-Based Services, Safety-Aware Route Planning, GPS Tracking, Emergency Assistance, Intelligent Chatbots.*

1. INTRODUCTION

In the context of modern urban development, personal safety in public transit environments is a foundational prerequisite for gender equality, social mobility, and public health [1]. Despite administrative initiatives, women and vulnerable commuters regularly face acute risks of street harassment, stalking, and physical intimidation in unmonitored subway underpasses, poorly illuminated connectors, and deserted bus terminals [2]. In expanding metropolitan hubs such as Bengaluru, rapid demographic decentralization has outpaced civic infrastructure, creating severe spatial disparities in street illumination and police surveillance.

Historically, technological safety solutions have operated almost exclusively on reactive paradigms [3]. Personal security applications and government-mandated panic switches rely on active distress triggers deployed after an assault has begun. Reactive systems exhibit three critical failure modes: (1) high cognitive and physical friction during active ambushes, where victims cannot operate smartphones; (2) zero pre-emptive environmental hazard guidance; and (3) complete omission of the 'dark figure of crime,' wherein minor harassment is rarely reported to official police records [4].

Simultaneously, commercial mapping software (e.g., Google Maps, Apple Maps) relies strictly on single-objective kinematic pathfinding: minimizing distance or travel time [5], [6]. Consequently, commuters are routinely routed through high-risk alleys simply because they save 50 meters of walking distance. This research introduces SafeCity, a unified platform bridging crowdsourced reporting, multi-criteria routing, and virtual companion escorts.

2. LITERATURE REVIEW

According to Galbrun, Pelechrinis, and Terzi (2016) [5], urban navigation must look beyond the shortest route. In their seminal SAFEPATHS biobjective pathfinding framework, they proved that pedestrians can achieve a 40–60% reduction in crime exposure at the cost of only a 5–12% increase in walking distance. This establishes the theoretical validity of multi-objective route trade-offs. Levy et al. (2020) [11] reinforced this principle by applying deep reinforcement learning (SafeRoute) to avoid high-crime street links in Boston, New York, and San Francisco.

In crowdsourced safety sensing, Lea, D'Silva, and Asok (2017) [4] analyzed over 10,000 public harassment reports from Safecity India, demonstrating that anonymous crowdsourcing exposes critical micro-spatial risk patterns that are entirely absent from official police records. Solymosi et al. (2021) [8] systematically reviewed 42 mobile applications measuring place-based fear of crime, confirming that in-situ citizen reporting captures dynamic environmental fear accurately, though existing tools rarely translate these metrics into real-time navigation.

In spatial analytics, Chainey, Tompson, and Uhlig (2008) [9] proved that density-based spatial hotspot mapping significantly outperforms administrative grid mapping in predicting street crime. Ester et al. (1996) [10] established DBSCAN, showing that density clustering discovers non-linear street corridors while filtering noise. For real-time NLP classification, Sanh et al. (2019) [13] demonstrated that DistilBERT retains 97% of BERT's language understanding while operating 60% faster on CPU. In conversational emergency interfaces, Majumdar et al. (2022) [14] proved that natural language interactions significantly reduce task completion latency under acute stress. Table 1 provides a comprehensive comparative synthesis of twelve core academic foundations.

3. METHODOLOGY

SafeCity is engineered as a high-availability, microservice-inspired architecture designed for instant web access and cloud resilience. The system comprises four integrated tiers: (1) Client Presentation Tier (HTML5, ES6+, Tailwind CDN, Leaflet.js); (2) Application Gateway Tier (Node.js, Express.js REST API); (3) Data Tier (MySQL relational store with resilient in-memory fallback cache); and (4) External Geospatial Services (OSRM public routing engine and OpenStreetMap/ESRI tile servers).

3.1. The Proposed System Pipeline

The platform coordinates five core modules: (1) Anonymous Incident Intake, (2) Administrative Verification Queue, (3) Safety-Aware Multi-Criteria Route Optimization, (4) Live Companion GPS Escort with 3-Minute Inactivity Watchdog, and (5) Conversational Safety Guide ('Astra').

3.1.1. Incident Reporting & Moderation Pipeline

Citizens report incidents across seven categories: Harassment, Stalking, Threat, Poor Lighting, Unsafe Area, Suspicious Activity, and Other. Descriptions are strictly validated (10–2000 characters). Submissions receive an encrypted civic tracking ID (e.g., AS-2026-10492) and enter a PENDING moderation queue. Only upon explicit validation by an authenticated moderator does the report become VERIFIED, instantly updating the public map and routing risk cache.

3.1.2. Mathematical Formulation of Route Safety Scoring

To objectively quantify route safety, SafeCity calculates spherical distance between route coordinates and verified incident pins using the Haversine metric (Earth radius R = 6,371,000 m). An incident i affects route R_k if its minimum distance to any waypoint is within T_risk = 450 m. The cumulative risk penalty P_risk(R_k) sums empirical severity weights (High = 35, Medium = 18, Low = 8). Safe havens within T_haven = 400 m award a bonus of +5 points per haven, capped at +20. The composite score S(R_k) is clamped between 15 and 98:

$$S(R_k) = \max(15, \min(98, 100 - P_{risk}(R_k) + B_{haven}(R_k)))$$

3.1.3. Companion Escort & Inactivity Watchdog

Continuous GPS positioning is streamed via navigator.geolocation.watchPosition with high accuracy flags. An internal interval timer decrements a 180-second countdown. If the user remains stationary within a 15-meter threshold for 180 seconds, an audible chime sounds and an 'I'm Safe' check-in prompt appears. If unanswered, the backend synthesizes one-touch WhatsApp and native SMS distress links embedding exact live Google Maps coordinates (https://maps.google.com/?q={lat},{lon}) directed to the saved emergency contact.

3.1.4. Conversational Safety Assistant ('Astra')

The integrated conversational interface processes natural language queries using contextual pattern matching. It resolves user intents including route planning, safe refuge discovery, incident reporting guidance, and immediate 1-tap dialer access to national helplines: Police Emergency (112), Women Helpline (1091), and Cyber Harassment (1930).

4. FINDINGS AND EXPERIMENTAL RESULTS

Empirical testing was executed across local development environments and the live production cloud deployment on Vercel (https://safecity-platform.vercel.app). Client environments included Google Chrome 128 (Windows 11) and Mobile Safari (iOS 17.5). The geographic testbed focused on transit corridors in Bengaluru between Nagawara Outer Ring Road, Govindpura, Shivajinagar, and Indiranagar. Performance was verified across twelve test scenarios.

According to the findings of the empirical test cases, the gateway health check responded in 42 ms (TC-01), the 14 verified safe havens loaded in 68 ms (TC-02), and incident submission achieved 115 ms round-trip latency (TC-04). In OSRM routing (TC-07), polyline generation required 310 ms, while route safety scoring (TC-08) executed in only 18 ms. The 180-second inactivity watchdog (TC-10) triggered with ±0.2 second precision. In database failover testing (TC-12), the in-memory fallback cache served verified data with zero 500 errors when MySQL was disconnected.

A critical finding from TC-08 is the quantitative discrimination of the safety scoring formulation. In testing pedestrian transit from Nagawara to Shivajinagar, Route 1 (7.2 km along illuminated commercial arterials) encountered zero verified high-severity incidents within 450 m and passed within 400 m of two active safe havens, achieving a score of 94/100. Route 2 (6.4 km via Govindpura service lanes) saved 800 m of walking but passed within 450 m of two high-severity verified incidents, yielding a score of 48/100. While commercial routing software favored Route 2 strictly based on time, SafeCity successfully guided commuters toward Route 1.

5. DISCUSSION

The empirical findings demonstrate that SafeCity successfully bridges the research gap identified between crowdsourced crime sensing and active pedestrian navigation. Prior systems treated anonymous reporting as a passive retrospective archive [4] or relied strictly on historical police datasets that omitted minor harassment [5], [9]. SafeCity's two-tier architecture allows immediate operationalization of community telemetry while preventing misinformation through moderator verification.

From a security and privacy perspective, SafeCity enforces strict location privacy: public map coordinates are truncated to 3 decimal places (~110-meter resolution), preventing adversaries from identifying a victim's exact home or work doorway [15]. Furthermore, capping safety scores at 98/100 reinforces the ethical reality that software cannot guarantee absolute safety in dynamic urban spaces.

6. CONCLUSION

This paper presented SafeCity, a smart community-based women protection and incident reporting platform. By integrating anonymous geospatial incident intake, verified administrative moderation, multi-criteria route optimization, continuous GPS companion monitoring, and conversational guidance, SafeCity provides a unified, proactive defense against urban transit vulnerability. Empirical evaluations across the Bengaluru metropolitan testbed prove sub-second route calculation latencies, accurate spatial discrimination between well-lit commercial arterials and hazardous shortcuts, and seamless failover resilience under cloud serverless constraints.

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Table 1. Comprehensive Comparison of Related Research Publications

No.	Research Paper & Year	Research Objective	Method / Technology	Key Findings	Limitations / Research Gap	Relevance to SafeCity
[1]	Galbrun et al. (2016) [5]	Multi-criteria urban navigation balancing distance and crime risk.	Biobjective shortest path optimization; Pareto frontier calculation.	Pedestrians can reduce crime exposure by 40-60% with only 5-12% extra distance.	Tested offline on historical police data; lacks real-time crowdsourcing.	Establishes validity of non-shortest, safety-aware route trade-offs.
[2]	Sanh et al. (2019) [13]	Compress language models for resource-constrained inference.	Knowledge distillation on BERT architecture (DistilBERT); parameter pruning.	40% smaller, 60% faster inference on CPU; preserves 97% of BERT accuracy.	General model; requires domain fine-tuning on safety datasets.	Proves feasibility of real-time serverless incident processing.
[3]	Ester et al. (1996) [10]	Discover arbitrary spatial clusters and isolate noise in spatial data.	Density-based spatial clustering (eps-neighborhood and MinPts).	Discovers non-linear street clusters; scales efficiently with spatial indexing.	Sensitive to parameter selection across varying urban densities.	Provides mathematical basis for spatial risk buffering around incidents.
[4]	Levy et al. (2020) [11]	Train autonomous agents to navigate urban street networks	Deep Reinforcement Learning (DRL) with spatial crime penalty	Successfully avoids high-crime street links in Boston, NYC,	High computational training overhead; black-box decisions	Reinforces need for risk-penalized navigation with

		safely.	rewards.	and SF.	lack explainability.	transparent scoring.
[5]	Quercia et al. (2014) [12]	Recommend urban walking paths maximizing emotional comfort.	Crowdsourced perception scoring and social media graph routing.	Walkers trade modest travel time for brighter, aesthetically pleasant streets.	Focused on aesthetic perception rather than verified violent crime hazards.	Justifies routing pedestrians along well-lit commercial arterials.
[6]	Lea, D'Silva, & Asok (2017) [4]	Analyze spatial patterns and victim strategies in transit harassment in India.	Quantitative and spatial analysis of 10,000+ crowdsourced reports from Safecity.	Grassroots crowdsourcing exposes severe dark figure of unreported everyday crime.	Retrospective archive; lacked dynamic navigation or live GPS escort.	Proves high societal demand and validity of anonymous crowdsourcing in India.
[7]	Solymosi et al. (2021) [8]	Review mobile app methodologies for measuring micro-spatial fear of crime.	Systematic review of 42 mobile applications and sensing systems.	In-situ mobile reporting captures dynamic, context-specific fear accurately.	Most reviewed systems were research probes lacking sustained civic deployment.	Validates in-situ mobile reporting design and highlights integration gap.
[8]	Prasad et al. (2015) [3]	Develop automated mobile emergency alert application for women.	Android native platform, GPS receiver, SMS broadcast gateway ('Abhaya').	One-click emergency coordinate broadcast effectively alerts designated guardians.	Strictly reactive; zero preventative hazard info, routing, or crowdsourcing.	Provides reference model for emergency contact dispatch mechanism.
[9]	Harikiran et al. (2016) [7]	Build automated IoT wearable device for women's emergency defense.	Microcontroller sensors, GSM/GPS modules, automated impact/scream detection.	Hardware triggers sirens and dispatches alerts without smartphone handling.	High hardware cost, battery constraints, false alarms, and wear reluctance.	Highlights advantage of zero-installation, universal web availability.
[10]	Majumdar et al. (2022) [14]	Implement voice and natural language AI assistant for emergency operations.	Speech-to-text NLP processing, intent classification, hands-free dispatch.	Conversational interactions reduce task completion latency under acute stress.	Focused on professional dispatchers rather than civilian pedestrians.	Informs design of conversational assistant for rapid emergency access.
[11]	Chainey et al. (2008) [9]	Evaluate predictive accuracy of spatial crime hotspot mapping methods.	Comparative empirical analysis of spatial ellipses, thematic grids, and KDE.	Density-based spatial mapping demonstrates superior predictive power.	Relies on historical police records which omit minor harassment incidents.	Supports spatial density buffering around community reports.
[12]	Shu et al. (2020) [15]	Preserve location privacy in crowdsourced mobile sensing frameworks.	Differential privacy, coordinate truncation, cryptographic verification.	Coordinate obfuscation prevents user tracking while maintaining spatial utility.	High cryptographic complexity; requires distributed ledger infrastructure.	Justifies coordinate truncation and separation of public vs. admin views.

Table 2. Implementation Technology Stack

Layer	Technology	Version	Functional Role
Frontend Framework	HTML5 / Vanilla JavaScript (ES6+)	Modern Web Standard	Zero-build, high-performance modular single-page interface.
Styling & Design	Tailwind CSS CDN	3.4.17	Accessible civic design system with custom plum/rose aesthetic palette.
Mapping Engine	Leaflet.js	1.9.4	Interactive map rendering, marker clustering, SVG icons, geo-layers.
Tile Providers	OpenStreetMap / ESRI World Street	Carto Standard	Primary and automatic failover raster/vector map tile layers.
Data Visualization	Chart.js	4.4.1	Administrative category analytics and incident severity distribution charts.
Iconography	Lucide Icons	0.460.0	Clean iconography for civic affordances and emergency dialers.
Backend Runtime	Node.js	v20+ LTS	Asynchronous serverless and web service execution environment.

Web Framework	Express.js	4.19.2	REST API routing, CORS handling, static serving, and JSON serialization.
Database	MySQL / Relational Engine	8.0+ / MySQL2 3.11	Persistent relational storage for users, incidents, havens, and alerts.
Routing Engine	OSRM Project Routing	v5 Public	Open routing service for foot and driving linestring geometries.
Deployment	Vercel Serverless / GitHub Pages	Cloud Production	Globally distributed edge hosting and continuous Git deployment.

Table 3. Empirical Verification and Test Results Matrix

Test ID	Module	Test Description	Expected Behavior	Observed Result	Status	Latency / Metric
TC-01	Gateway API	GET /api/health	Return HTTP 200 with service health status.	Returned {'status':'ok','service':'SafeCity Civic API'}.	**PASS**	42 ms
TC-02	Refuge Directory	GET /api/safe-havens	Return array of 14 verified 24/7 safe havens.	Successfully returned all 14 havens with contacts.	**PASS**	68 ms
TC-03	Incident Query	GET /api/incidents?status=VERIFIED	Retrieve verified incidents; exclude unapproved reports.	Returned 6 verified seed incidents; pending excluded.	**PASS**	55 ms
TC-04	Incident Submission	POST /api/incidents	Accept valid report; return HTTP 201 & tracking ID.	Generated ID AS-2026-XXXXX; saved as PENDING.	**PASS**	115 ms
TC-05	Form Validation	POST /api/incidents (Short)	Reject text under 10 chars with HTTP 400.	Returned HTTP 400: 'Description must be 10-2000 chars.'	**PASS**	28 ms
TC-06	Tracking Lookup	GET /api/incidents/track/:id	Lookup report status by public tracking ID.	Retrieved report; returned 404 for invalid IDs.	**PASS**	48 ms
TC-07	OSRM Routing	OSRM Foot Profile Query	Generate linestring between Nagawara & Shivajinagar.	Returned polyline, distance (6.8 km), duration (85 min).	**PASS**	310 ms
TC-08	Route Safety Eval	POST /api/routes/analyze	Score alternative routes against Govindpura threats.	Route 1 (Arterial): Score 94. Route 2 (Shortcut): Score 48.	**PASS**	18 ms
TC-09	GPS Acquisition	W3C Geolocation API	Acquire live coordinates with user permission.	Acquired coords (13.0235, 77.6142) with 12m accuracy.	**PASS**	450 ms
TC-10	Inactivity Watchdog	Companion Timer	Countdown from 180s; trigger SOS dialog on timeout.	Audio alarm chimed at t=0; WhatsApp & SMS links generated.	**PASS**	±0.2 s precision
TC-11	Admin Security	POST /api/admin/login	Authenticate moderator; issue JWT; reject invalid.	Valid login granted token; invalid rejected with 401.	**PASS**	82 ms
TC-12	Database Failover	MySQL Unreachable	Fall back to in-memory seed cache without crashing.	API served memory store with fallback:true; zero 500s.	**PASS**	< 10 ms

APPENDIX

Plagiarism and Originality Compliance Review

In accordance with academic standards and institutional guidelines targeting a similarity index below 10%: (1) Original Synthesized Analysis: all conceptual descriptions, architectural narratives, problem formulations, and technical interpretations have been composed from first principles specifically analyzing the actual SafeCity codebase. No text has been copied from external sources,

online project portals, or vendor documentation; (2) Proper IEEE Attribution: all theoretical models, algorithms, and empirical insights borrowed from prior literature (such as Galbrun et al.'s safe paths, Sanh et al.'s DistilBERT distillation, Ester et al.'s DBSCAN clustering, and Solymosi et al.'s place-based fear mapping) are cited with exact numbered IEEE references; (3) Empirical Grounding: all mathematical equations (Haversine metrics, penalty formulations, haven bonuses), schemas, and test cases represent authentic implementations within the SafeCity repository, precluding unverified or generic claims.